

2022-Ph-H1-Q4

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

Which velocity–time graph shows uniform acceleration?

Answer: E

Explanation:

Uniform acceleration means the velocity changes at a constant rate — this gives a **straight line** on a velocity–time graph, **not horizontal or curved**.

Guidance for Students:

Recognise graph shapes:

- Straight line = constant acceleration
- Curve = changing acceleration
- Horizontal = constant velocity

Revision Tips:

- v – t straight line $\neq$ constant velocity!
- Gradient of v – t = acceleration
- Acceleration = change in velocity / time